

Kakoli Rani Ray Koli

📍 Sylhet, Bangladesh 📞 01300587091 ✉ rkakoli333@gmail.com

📁 Portfolio 🔗 LinkedIn 🎓 Google Scholar 🐙 GitHub

Professional Summary

Computer Science and Engineering graduate with a CGPA of **3.97/4.00** with distinction, ranked **1st** in the department. Published **two IEEE conference papers** and conducted research in Machine Learning, Deep Learning, Computer Vision, NLP, and Healthcare AI. Strong foundation in algorithms and programming, with approximately 1000 solved competitive programming problems. Seeking a Lecturer/Faculty position to contribute to undergraduate teaching, research, student supervision, and academic development.

Education

B.Sc.(Engg.) in Computer Science and Engineering

Feb 2022 – Sep 2026

Sylhet Engineering College, Sylhet, Bangladesh

CGPA: 3.97/4.00 (1st Position in Department)

Higher Secondary Certificate (HSC), Science

Jul 2018 – Aug 2020

Sunamganj Govt. College, Sunamganj, Bangladesh

GPA: 5.00/5.00 (Board: Sylhet)

Research Experience

Undergraduate Researcher

2025–2026

Supervisor: Md Abu Naser Majumder, Associate Professor, Department of CSE, Sylhet Engineering College

- **CAPT: Exploring the Performance–Efficiency Trade-off in Bangla Medical Symptom-to-Specialist Classification**
 - Developed a lightweight model using 7,498 Bangla symptom sentences across 12 classes.
 - Achieved 92.18% accuracy with approximately 330K parameters and a model size of about 1.26 MB, outperforming several lightweight baselines.
- **tomato Leaf Disease Detection Using Modified Yolo11**
 - Curated a dataset of 1,917 tomato leaf images across 8 disease classes.
 - Developing a Modified Yolo11 model for accurate and resource-efficient multi-class detection.

Publications

Published -

1. Md. Ragib Hasan, **Kakoli Rani Ray Koli**, Naimur Rahman, “ChakmaCNN: A Pronunciation-Aware CNN for Handwritten Chakma Script Recognition,” *2025 28th International Conference on Computer and Information Technology (ICCIT)*, IEEE, 2025. **2nd Author**.
DOI: 10.1109/ICCIT68739.2025.11491110
2. Sandwip Dash Talukder, **Kakoli Rani Ray Koli**, Ayan Banik, Bishal Roy, “Bridging Tradition and Technology: Automated Classification of Bengali Folk Music Using Machine Learning,” *2025 28th International Conference on Computer and Information Technology (ICCIT)*, IEEE, 2025. **2nd Author**.
DOI: 10.1109/ICCIT68739.2025.11491301

Ongoing and Under Review -

1. **Kakoli Rani Ray Koli**, Ayan Banik, Saif Ahmed, “EfficientBrainNet: A Lightweight CNN for Brain Tumor MRI Classification,” **1st Author**.
Status: Submitted in Conference
2. Ayan Banik, **Kakoli Rani Ray Koli**, Saif Ahmed, “Lightweight Deep Learning for Automated Multi-Class Shoe Categorization in Online Shopping Platforms,” **2nd Author**.
Status: Ongoing
3. Arif Ahmed, **Kakoli Rani Ray Koli**, Ayan Banik, Nafiz Nahid, Md Abu Naser Majumder “Tomato Leaf Disease Detection Using Modified Yolo11,” **2nd Author**.
Status: Ongoing

Research Interests

Machine Learning • Deep Learning • Computer Vision • NLP • Healthcare AI • Data Science • Data Analysis

Projects

MediScan AI



AI-Based Multi-Disease Medical Image Classification Web Application

Technologies: Deep Learning, Streamlit, HTML, CSS, JavaScript

- Developed a medical image classification system for brain tumor, kidney stone, lung disease, and eye disease prediction using separately trained deep learning models.
- Integrated all four models into a single interactive web application for image upload and prediction using Streamlit, HTML, CSS, and JavaScript.

Cafe Management System



Desktop Application for Cafe Order and Billing Management

Technologies: Java, Java Swing, Object-Oriented Programming

- Developed a desktop-based cafe management application for handling customer orders and billing.
- Implemented food selection, quantity adjustment, item removal, order reset, and printable bill generation using Java Swing.

Technical Skills

Programming:	C, C++, Python, Java
AI/ML:	Scikit-learn, TensorFlow
Data Analysis & Visualization:	Pandas, NumPy, Matplotlib, Seaborn
Research & Documentation:	Latex, Overleaf
Web Technologies:	HTML, CSS, JavaScript, Flask, Streamlit
Tools & Development:	Git, GitHub, Jupyter Notebook, Google Colab, VS Code

Awards & Achievements

- **Champion** : Satellite Data Analytics with Power BI - Climate & Environmental Intelligence, Organized By - Metropolitan University Geography and Astronomical Society
- **Codeforces**: Maximum Rating **1208 (Pupil)** - Profile: kakoli
- **CodeChef**: Maximum Rating **1547 (2★)** - Profile: kakoli
- Solved approximately **1000 problems** across multiple online judges.

Academic & Research Activities

3rd Student Research Conference, Leading University

- Presented research at the 3rd Student Research Conference, Leading University.

Leadership & Responsibilities

Female Organizing Secretary , CSE Research Club, Sylhet Engineering College	Nov 2025 – Sep 2026
Female Organizing Secretary , SEC Programming Club	Nov 2025 – Sep 2026
Joint Secretary , SEC CSE Society	Nov 2025 – Sep 2026

Languages

Bengali: Native

English: Fluent

References

1. **Md Abu Naser Majumder** – Associate Professor & Head
Department of Computer Science & Engineering, Sylhet Engineering College
Thesis Supervisor, Course Instructor
☎ 01712291794
✉ abu.naser84@gmail.com
2. **Md Lysuzzaman** – Assistant Professor
Department of Computer Science & Engineering, Mymensingh Engineering College
Course Instructor
☎ 01719340955
✉ lzaman.cse@mec.ac.bd